

# Harsh Shukla

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## EDUCATION

### McMaster University

Bachelor of Engineering in Software Engineering (Co-op), GPA: 3.9/4.0

Hamilton, ON

Sep 2023 – Apr 2028 (Expected)

## EXPERIENCE

### Software Developer, Exoskeleton Team

McMaster University

Dec 2024 – Present

Hamilton, ON

- Engineered real-time data pipeline for 10+ inertial sensors using C++ and ROS, reducing latency by 60% and enabling adaptive gait control loops.
- Implemented extended Kalman filter for multi-sensor fusion, improving sensor accuracy by **35%** and stability during dynamic movements.
- Collaborated with cross-functional team to integrate AI-based **gait prediction models** in **Python**, achieving **92%** accuracy in stride phase detection.

### Full-Stack Web Developer

Site4U Foundation

May 2024 – Nov 2024

Ontario, Canada

- Designed and built **5+** responsive **React** applications with **Vite** and **Tailwind**, increasing client engagement metrics by **50%**.
- Migrated legacy sites to **React** and **Firebase**, reducing page load times by **40%** and improving **SEO** rankings.
- Automated **deployment** workflows with **GitHub Actions** & **Azure**, cutting release cycle from 2 days to 2 hours.

### Co-Captain & Lead Developer

10700X VEX Robotics Team

Oct 2022 – Jun 2023

Brampton, ON

- Founded and led a 12-member robotics team, securing **\$8K** in sponsorships and achieving **3rd** place out of **64** teams provincially.
- Implemented **C++** autonomous routines with state-machine logic, improving match win-rate by **60%**.
- Mentored junior programmers in **Git workflows** and **C++** best practices, reducing onboarding time by **50%**.

## PROJECTS

### GryphHacks Volunteering Forum

May 2022

- Built full-stack web app with **React**, **Node.js**, and **Firebase** to connect volunteers with events, managing event listings.
- Integrated **Leaflet.js** for geospatial event discovery and implemented secure **RESTful APIs** in **C++** backend.
- Delivered MVP in 48 hours, placing **2nd** among **70+** international teams.

### EcoHub Rewards Platform

Aug 2024

- Developed **React** frontend and **Firebase/Firestore** backend for eco-activity tracking, increasing user participation by **30%**.
- Implemented blockchain-inspired EcoCoin system on **Google Cloud Functions**, handling transactions securely.

### CyberSecurity System

Nov 2024

- Designed a comprehensive system integrating **facial recognition**, a password manager, and file **encryption** capabilities using real time video and symmetric key encryption
- Leveraged technologies including **Python** (Flask, OpenCV, Cryptography), and **SQL** to ensure robust functionality with less than **2ms** response time
- Achieved 2nd place among **30+** teams, recognized for technical innovation and practical execution

### Rigor Fitness Tracker

Sep 2023

- Created **React/Firebase** web app with **Google Vision** API for meal logging, achieving **85% OCR** accuracy.
- Implemented user authentication and analytics dashboard to track macronutrient goals

## TECHNICAL SKILLS

**Languages:** Python, C++, Java, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

**Frameworks/Tools:** React, Node.js, Docker, Kubernetes, Firebase, Azure, ROS, Git, GitHub Actions, Jenkins, Postman, UML

**Concepts:** Data Structures & Algorithms, OOP, RESTful APIs, Microservices, Agile/Scrum, CI/CD